AdS GNN: SuperPixel MNIST

- We test it on **SuperPixel MNIST** (Monti 2017), a dataset where each entry has a point cloud of 75 pixels and the network has to find the digit.

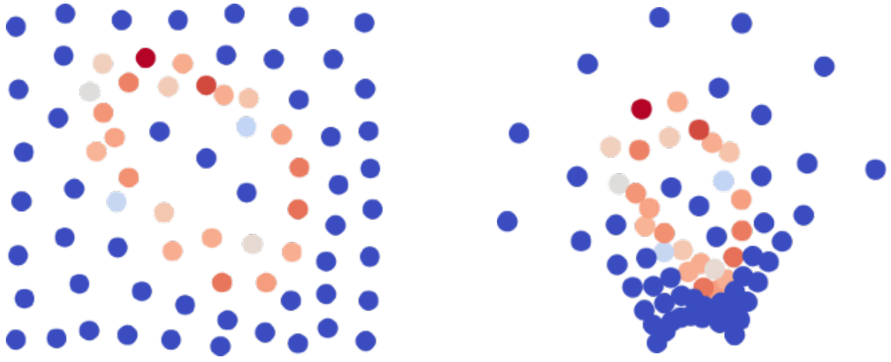
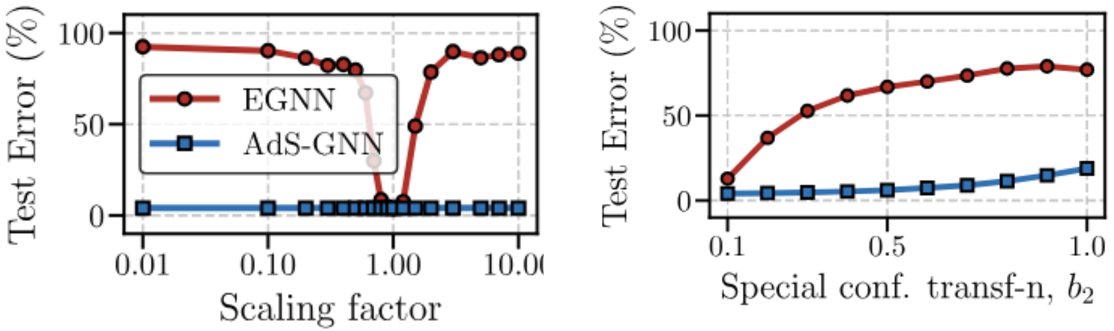


Table 1: Classification error on SuperPixel MNIST. We augment the test set with random rotations and scaling. ✗ indicates random-guess performance.

Model	Error rate, %		
	non-augmented	rotated	rotated+scaled
MONET	8.89	✗	✗
SplineCNN	4.78	✗	✗
GCGP	4.2	✗	✗
GAT	3.81	✗	✗
PNCNN	1.24 ± 0.12	✗	✗
PΘNITA	1.17 ± 0.11	1.17	✗
EGNN	4.17 ± 0.45	4.17	✗
AdS-GNN (**Ours**)	4.09 ± 0.27	4.09	4.09

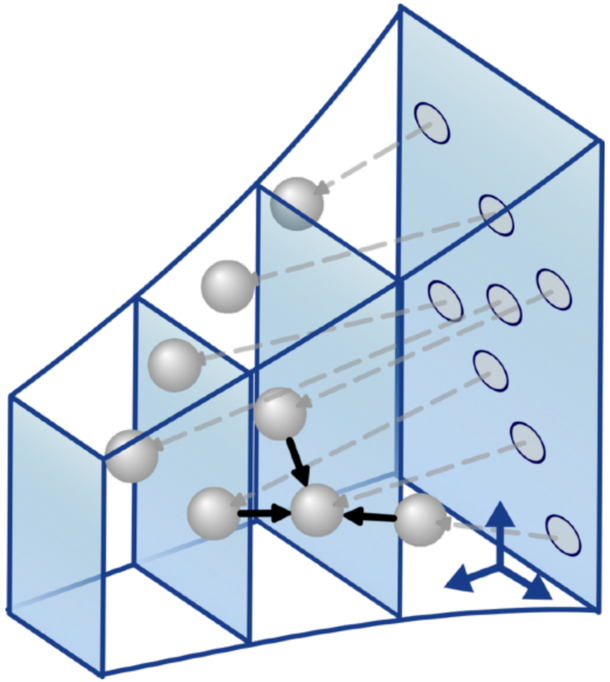
Ours is comparable to existing ones, but exactly invariant under scaling.

Figure 2: Test error on augmented data, SuperPixel MNIST.



Why AdS?

- Convenient way to capture the **physically relevant part** of conformally invariant data.
- Conditioning on AdS proper distance naturally introduces **notion of locality** in scale.
- Idea of **"scale space"** has long history in computer vision — our use of AdS is a special version of scale space that also enforces special conformal transformation.
- This is an example of how ideas from a different field (i.e. string theory) allowed for a **genuinely novel solution** of an existing problem!



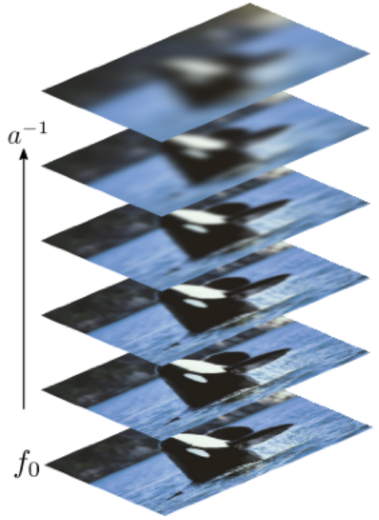


Figure 2: A Scale-space: For implementations we logarithmically discretize the scale-axis in a^{-1}.

Worrall & Welling 2019